SC2107
Microprocessor System Design and Dev

Lab5 Assessment
(Updated 17 Oct 2024)

Course Assessment

- 100% Continuous Assessment
  - No Semester-end written examination
- Continuous Assessment comprises of:
  - **Two Quizzes** (50%)
    - Open Book
    - Week 8 and Week 13 (Details will be announced later)
  - **Lab Assignments and Assessment** (50%)
    - **Four Lab Assignments (20%)**
      - Lab 1, 2, 3, 4
      - Handout to be filled during lab session.
      - Submitted in softcopy through the Assignment folder in NTU Learn before next lab.
    - **One Practical Test (30%)**

Makeup Lab

- Makeup Labs (Lab1-4)
  - Approved Makeup must be done before next Lab, else will be void and zero marks given for assignment.
  - Due date of assignment remains unchanged (date of originally scheduled next lab).
  - Only one makeup allowed for each lab.

- Makeup Lab (Lab5)
  - Approved Makeup must be done on allocated date.
  - Only one makeup allowed.
  - Penalty marks will be applied as students has more time to prepare.
  - All makeups must be completed by week 14.

Lab5 Practical Test (30% overall grade)

- Be punctual.
- No communications between students. Disciplinary action will be sanctioned to students caught talking to each other.
- Assessment includes
  - Assigned mini tasks to be completed within Lab session.
- All assessment must be completed within Lab5 session. Assessment not cleared by end of lab session will have zero marks for the corresponding items. See additional time allocation for each task in following slides.
- Each student will be provided a RSLK Max hardware during Lab5

Lab5 Assigned Mini Tasks

- Pre-known coding tasks involving work done in Lab2-4 (20% marks).
- Two coding tasks from task list (2L, 1M, 1H) released during Lab5. Students can choose one of the following options.
  - Option A: 2x L-tasks (20% each, Total 40%)
  - Option B: 1x L-task (20%) and 1x M-task (35%)
  - Option C: 1x L-task (20%) and 1x H-task (50%)
  - See following slides for examples of L, M and H tasks.
- Time allocation for assessment
  - Pre-known task needs to complete within 1 hour from start of Lab5.
  - Option A and B needs to complete within 2 hours.
  - Option C needs to complete within 2.5 hours. If student can complete within 2 hours, 10% bonus marks awarded.
  - All timings reference official start time of lab5, no additional time given for late comers.
- Inform supervisors/TA upon completion of each task for their work to be validated. Do not attempt to validate all tasks at one go.
- Multiple attempts for each tasks are allowed. No penalty if student failed to show the correct result.

Lab5 Pre-known task details

- Lab2
  - Run the Lab2 application, place each of the reflectance sensor above a black tape on the table, the output of the Reflectance_Read() function in your code should show the corresponding output. You can show the result in CCS watch window.
- Lab3
  - Run the Lab3 application, press the Bump switches one at a time. Your application should show which Bump switch is being pressed. All six bump switches must be detected correctly. You can show the result in CCS watch window.
- Lab4
  - Run the Lab4 application, place an obstacle 20-30cm away from each of the L, C and R IR sensor. Your application should show the distance correctly, +/- 5cm accuracy. All three IR sensors must sense the correct distance. You can show the result in CCS watch window.

Complete these tasks before coming to the lab!

Lab5 Assigned Mini Tasks Examples

- L-task.
  - Application of one task done in lab1-4
  - Example: Blink the RED LED only when black line is below reflectance sensor #1.
- M-task.
  - Application involving interaction of two tasks in lab1-4
  - Example: Move the RSLK, turn the RSLK 90 degree to the right when reflectance sensor #1 detected a black line.
- H-task.
  - Application involving simple design and/or algor
  - Example: Implement a binary to decimal converter using input from the bump switches (GPIO-based interrupt). Bump0=bit0, Bump1=bit1…, Bump5=bit5. Display the converted decimal and hexadecimal number on a Serial Terminal. Use of conversion library not allowed.
  - Example: Enable RSLK to move from start point to end point of maze #1. Ensure RSLK is parked properly into the carpark at the end point.

Sample Task Slide (Projected on Screen in Lab)

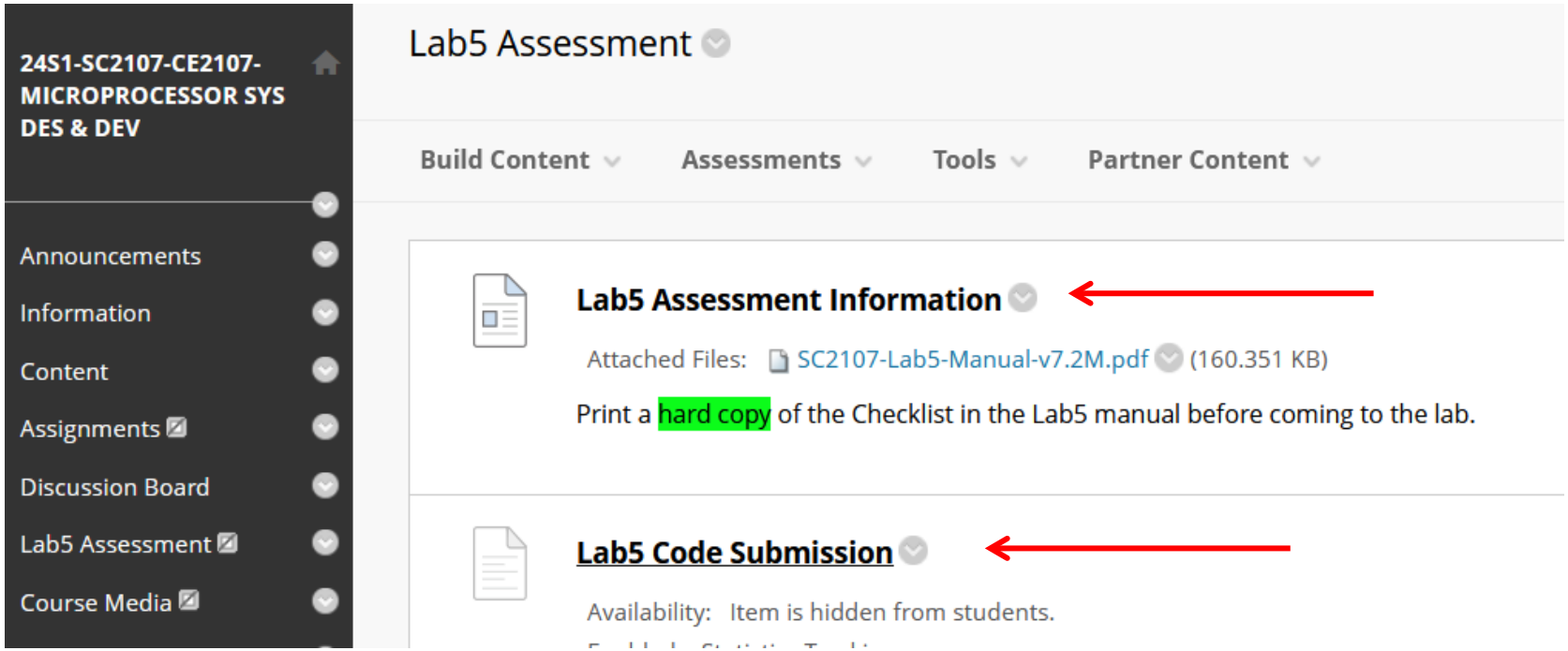
Lab Group	Task	Description	Assessment
LAB_GRP	L1	Blink the RED LED only when black line is below reflectance sensor #1	Place the RSLK such that reflectance sensor#1 is above the black tape, show that the RED LED is blinking. Note that the LED needs to be seen by human eye to be blinking.
	L2	L-Task2	Steps to evaluate
	M	Move the RSLK, turn the RSLK 90 degree to the right when reflectance sensor #1 detected a black line.	Place the RSLK robot a distance away from a black line. RSLK should move forward and stop when it hits the black line.
	H	Implement a binary to decimal converter using input from the bump switches (GPIO-based interrupt). Bump0=bit0, Bump1=bit1..., Bump5=bit5. Display the converted decimal and hexadecimal number on a Serial Terminal. Use of conversion library not allowed.	Press one of more bump switches and verify that the correct dec and hex value is displayed on the serial terminal. Ensure all bump switches are tested. Check that GPIO based interrupt is used.

Lab5 Practical Test

- No discussion during lab5. Once you are done the task assessment, please leave the lab quietly.
- Phones are not allowed, please keep them in the bag.
- Raise your hand when you are ready for Task review.
- Hand in the assessment sheet before leaving the lab. The sheet will be voided if you bring it out of the lab.
- Upload zipped version of your task code to NTULearn “Lab5 Assessment” folder
  - Filename Format: Fullname.zip e.g. AayBeeCee.zip
  - Check to ensure the zip file is uploaded successfully. Retain any evidence to support your claim if there are any discrepancies later.
  - This source will be used to validate the demo you show during the lab5 Practical Test.

NTULearn

- Lab5 Practical Test Information and
- Code Submission (link visible nearer to Lab5)



Important Points to note

- Same rules as Quiz. Some key rules below. Check Quiz doc for details.
  - Keep mobile phone away.
  - No communication between students during lab5.
  - Open Book, can use google but no GenAI tools.
- You can use your lab top for your coding during lab5.
- If using lap PC, make sure your process of transferring your code to the lab PC has been tried and tested.
- DO NOT execute your code directly from USB flash drives. Doing so tend to lead to system environment related issues e.g. CCS failed to find source file for compilation.
- DO NOT execute your code directly from compressed file. Unzip the file before using.